# ANIKATE CHOWDHURY

**Geologist | Mineral Exploration | Remote Sensing & GIS | Geological Modelling**

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## PROFESSIONAL SUMMARY

Geologist (M.Sc. Applied Geology, 2026) with hands-on exploration experience at MMPL Pvt. Ltd., preparing G3/G4 exploration proposals and MEMC Schedule-I geological reports, drawing AutoCAD cross-sections and borehole correlations, and applying multispectral, SAR and hyperspectral remote sensing to prospectivity mapping. Trained at the Geological Survey of India, Atomic Minerals Directorate and Indian Bureau of Mines in critical-mineral exploration, core logging and resource estimation, with a strong foundation in hydrogeology, geophysics and Python-based machine learning. Open to relocation for exploration, mining and geospatial roles.

## PROFESSIONAL EXPERIENCE

**Intern-Geologist** Aug 2026 – Present

*MMPL Pvt. Ltd. (formerly Maheshwari Mining Pvt. Ltd.), Bhopal, Madhya Pradesh | On-site*

- Prepared two G4 base-metal exploration proposals (Chhatarpur, Shivpuri) and one G3 iron ore proposal (Singrauli), integrating geological information, historical exploration data and prospectivity assessments into exploration programmes.
- Applied remote sensing and GIS (multispectral, SAR, hyperspectral) to prospectivity mapping, structural interpretation, lineament extraction and hydrothermal alteration mapping; integrated results with geological and historical data to delineate prospective zones and prioritise targets.
- Prepared two G4 limestone geological reports for Panna district per MEMC Schedule-I, integrating borehole logs, DGPS survey data, assay and laboratory results, and estimating mineral resources using graphical methods and geological cross-sections.
- Developed geological cross-sections and correlation sections in AutoCAD and performed borehole correlations to interpret subsurface lithological continuity, including for ongoing exploration projects.
- Planned scout borehole programmes and prepared drilling schedules based on geological interpretation, target locations and exploration objectives.
- Monitored exploration activity (geological mapping, sampling, pitting, drilling) across 8+ Central Zone blocks; prepared block-wise technical progress reports and management review documentation.
- Managed sample dispatch, laboratory advisories and analytical tracking; coordinated field logistics, equipment deployment, site communications, consultant billing, expenditure records and project documentation.

## TRAINING & FIELD EXPERIENCE

**Industrial and specialist training:** Refresher Course on Exploration for Critical & Strategic Minerals, Geological Survey of India, FTC Zawar (Sep 2026), covering REE, rare-metal, W, Sn, Li and Nb-Ta systems, field visits to lithium, tungsten and carbonatite prospects, core logging, MEMC rules, and resource estimation by cross-sectional and longitudinal vertical section methods. Atomic Minerals Directorate, Ghatshila (Aug 2025): geological mapping and radiometric surveying in the Surda–Terenga area of the Singhbhum Shear Zone, plus a visit to the Narwapahar underground mine. Indian Bureau of Mines, Nagpur (Jun 2025): mineral processing, ore characterization and chemical analysis (XRD, XRF, SEM, ICP-MS, TG-DTA exposure).

**Field work and research training:** Structural mapping and tectonic study along the North Purulia Shear Zone (2025); litho-structural mapping at Galudih, sedimentological study at Chandipur and petrological mapping at Bangriposi; Sukinda chromite mine visit; water-quality, bathymetric and biomass survey at Taki with Adamas University (2025); academic internships in hydrogeological (2023) and hydrogeochemical (2024) data analysis at Presidency University, Kolkata.

## TECHNICAL SKILLS

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| **Exploration & geology** | Prospectivity mapping, REE and critical minerals, hydrothermal alteration mapping, borehole planning and correlation, core logging, geological cross-sections, resource estimation (CS and LVS methods), MEMC/NMET reporting, handheld XRF, petrography and thin-section preparation |
| **GIS & remote sensing** | ArcGIS, QGIS, ENVI, Google Earth Engine, Google Earth Pro; multispectral, SAR and hyperspectral data analysis |
| **Geophysics & hydrogeology** | pyGIMLi (ERT inversion), VES, MODFLOW/MODPATH (GMS), HEC-RAS, PHREEQC, GRACE TWSA, Groundwater modelling, Geostatistics |
| **Data & machine learning** | Python (Pandas, NumPy, Scikit-learn, Matplotlib), machine learning, statistical data analysis |
| **Software & instruments** | AutoCAD, LibreCAD, Surfer,Strater, Grapher, RockWorks, HighScore Plus, Geo-Rose, Geo-Orient, CorelDRAW, Photoshop, MS Office; GPS (DNRGarmin), Abney level, echo sounder |

## EDUCATION

**M.Sc. Applied Geology** Sep 2024 – Jul 2026

*Presidency University, Kolkata | CGPA 8.65 / 10*

**Dissertation (submitted Mar 2026):** Multi-Criteria Based Geospatial and Computational Modelling for Groundwater Suitability Assessment in Urban Aquifer Systems: A Case Study from the Northern Yamuna Floodplain, NCT Delhi. Supervisor: Dr. Aditya Sarkar.

**B.Sc. (Hons) Geology** Sep 2021 – Aug 2024

*Presidency University, Kolkata | CGPA 7.5 / 10*

*DST-INSPIRE SHE Scholar, Government of India (2021).*

## PROJECTS

- **Drillhole cross-section construction and ore reserve estimation** (Python) – GitHub
- **2D ERT inversion and subsurface imaging** (pyGIMLi) – GitHub
- **Lithology and geochemical facies classification** (Random Forest, scikit-learn) – GitHub
- **ML-based mineral prospectivity prediction** (geological and structural evidence layers) – GitHub
- **Geochemical data exploration and statistical analysis** (Python) – GitHub
- **Well log petrophysical analysis and synthetic seismogram generation** (Python) – GitHub
- **Environmental impact assessment of mining-zone water quality** (geochemical indices, ML), collaborative – GitHub
- **Machine-learning-assisted Groundwater Resilience Index** (ML), collaborative – GitHub

## PUBLICATIONS & CONFERENCES

- **Evaluating Groundwater Resilience for WASH Planning in a Semi-Arid Region Through a GIS-Machine Learning Framework: Alwar District, Rajasthan** – National WASH Conference, SPM-NIWAS, 2026
- **Integrated Morphotectonic Assessment of the Torsa–Amo Chu River Basin Using Geomorphic Indices and Machine Learning** – BMES-2026, IIT (ISM) Dhanbad, 2026 (first author)
- **Machine Learning-Enhanced Analysis of Temporal b-Value Variations Across Different Tectonic Settings** – BMES-2026, IIT (ISM) Dhanbad, 2026
- **Identifying Groundwater Potential Zones Using GIS-Based Overlay Model in Parts of West Bengal: A Review Article** – BHU-VIDYA, 65th Edition, ISSN 0970-0323, 2025 (sole author)
- **Forecasting Groundwater Dynamics in Jalpaiguri District, North Bengal, Using Machine Learning and GRACE Satellite Data** – Water Summit 2025, Manav Rachna University (first author)
- **Integrated Hydro-Geophysical and GIS-Based Characterization of Micro-Level Aquifers in the Sundarbans, West Bengal** – Water Summit 2025, Manav Rachna University
- **Quantification of Groundwater Quality Issues in Reference to Urbanization in Parts of Barddhaman District, West Bengal** – 32nd West Bengal State Science & Technology Congress, 2024
- **Impact of Human Encroachment and Regional Meteorological Changes on Water Quality Around Nainital, Uttarakhand** – Dept. of Geology, Presidency University, Kolkata, 2024
- **Waste Water Management in Kolkata: Hydrological Constraints and Development of Urban Environment** – ISSWM-2023, IEM Kolkata, 2023

## CERTIFICATIONS & WORKSHOPS

- **Geological Survey of India (GSITI):** Regional Geophysical Mapping for Mineral Targeting (2026); Application of Remote Sensing in Mineral Exploration (2025); Workshop on Interpretation of Geophysical Datasets (2025)
- **IIRS, ISRO:** Remote Sensing and Digital Image Analysis (2025); Advances in Remote Sensing Techniques for Geological Applications (2025); Advanced Image Analysis for Geospatial Professionals (2025)
- **University of Oulu, Finland:** Application of Remote Sensing in Water Resources and Climate Change, PhD short course, 2 ECTS (2025)
- **Esri / Quest GIS:** ArcGIS Imagery MOOC: Foundations and Frontiers (2025); QGIS Mastery: Basic and Application (2025)
- **Python and ML:** Machine Learning I, Columbia University; Applied Data Science with Python, IBM (2025); Python Programming in Geology, TerraPy (2025); Python, Kaggle (2023)
- **IIT (ISM) Dhanbad:** Workshop: Bridging Mathematics and Earth Sciences, BMES-2026 (participation and poster)